NCERT • GRADE 7 • MATHS • PART 2

Finding the Unknown

C07 • Ganita Prakash Part II

Balance an equation • Solve step by step

Form equations • Check answers

Illustrated NCERT questions and worked solutions

Textbook solutions first • Visual understanding • Revision

Original student notes based on NCERT. All illustrations are editable LaTeX drawings.

Part A • NCERT questions and worked solutions

How this answer section is organised

This chapter includes three numbered *Figure it Out* groups (textbook pp. 172, 181 and 185–189), a nine-part *Mind the Mistake* exercise (pp. 181–182), and visual activities with weighing scales and matchsticks. Each answer below follows the **NCERT question order**. Question statements are summarised in new words so you can use the textbook alongside these notes.

Set 1 • Figure it Out, textbook page 172

Q1 • Solve and verify five equations

Part	Steps	Solution
(a) $3x - 10 = 35$	$3x = 45$	$\boxed{x = 15}$
(b) $5s = 3s$	$2s = 0$	$\boxed{s = 0}$
(c) $3u - 7 = 2u + 3$	$u - 7 = 3$	$\boxed{u = 10}$
(d) $4(m + 6) - 8 = 2m - 4$	$4m + 16 = 2m - 4;\ 2m = -20$	$\boxed{m = -10}$
(e) $u/15 = 6$	$u = 6 \times 15$	$\boxed{u = 90}$

Substitute the proposed number into both sides of the corresponding equation to confirm equality. For (d), $4(-10 + 6) - 8 = -24 = 2(-10) - 4$.

Q2 • Create an impossible equation

Example: $\boxed{x + 4 = x + 5}$. Subtract x from both sides to obtain $4 = 5$, which is impossible. This equation has **no solution**.

Set 2 • Figure it Out, textbook page 181

Q1 • Five different equations solved by $x = -2$

$x + 5 = 3$, $2x = -4$, $3x + 8 = 2$, $5x - 3 = -13$, $4(x + 3) = 4$. In every equation, replacing x by -2 makes LHS equal RHS.

Q2 • Solve all seven letter-number equations

Part	Work	Answer
(a) $2y = 60$	$y = 60/2$	$\boxed{30}$
(b) $-8 = 5x - 3$	$5x = -5$	$\boxed{-1}$
(c) $-53w = -15$	$w = (-15)/(-53)$	$\boxed{15/53}$
(d) $13 - z = 8$	$-z = -5$	$\boxed{5}$
(e) $k + 8 = 12 - k$	$2k = 4$	$\boxed{2}$
(f) $7m = m - 3$	$6m = -3$	$\boxed{-1/2}$
(g) $3n = 10 + n$	$2n = 10$	$\boxed{5}$

Important: (c) has the coefficient *negative fifty-three*, not a stacked fraction. Do not turn $-53w$ into $-5w/3$.

Q3 • The number with three related digits

Let the hundreds digit be a. Tens $= a+3$ and units $= a+6$. Their sum is $3a+9=15$, giving $a=2$. Digits: $2, 5, 8$. Answer $\boxed{258}$.

Q4 • Brick heavier by 1 kg than half itself

Let the brick weigh b kg. $b = b/2 + 1$, so $b/2 = 1$, hence $\boxed{b = 2 \text{ kg}}$.

Q5 • Quarter of a number plus nine

Let $n/4 + 9 = n$. Multiply by four: $n + 36 = 4n$. Thus $3n = 36$ and $\boxed{n = 12}$.

Q6 • Use $4k + 1 = 13$ without starting over

First $4k = 12$ and $k = 3$. (a) $8k + 2 = \boxed{26}$; (b) $4k = \boxed{12}$; (c) $k = \boxed{3}$; (d) $4k - 1 = \boxed{11}$; (e) $-k - 2 = \boxed{-5}$.

Set 3 • Mind the Mistake, textbook pages 181–182

Q1–Q3 • Fix the incorrect operations

1. $4x + 6 = 10$: subtract 6, not add it. $4x = 4 \Rightarrow \boxed{x = 1}$.

2. $7 - 8z = 5$: $8z = 2 \Rightarrow \boxed{z = \frac{1}{4}}$, not 4.

3. $2v - 4 = 6$: add 4 first: $2v = 10 \Rightarrow \boxed{v = 5}$. Do not divide only the first term of a sum.

Q4–Q6 • Correct the signs and terms

4. $5z + 2 = 3z - 4 \Rightarrow 2z = -6 \Rightarrow \boxed{z = -3}$. Subtract $3z$, do not add it.

5. $15w - 4w = 26 \Rightarrow 11w = 26 \Rightarrow \boxed{w = 26/11}$. Unlike terms with a variable, the 26 remains constant.

6. $3x + 1 = -12 \Rightarrow 3x = -13 \Rightarrow \boxed{x = -13/3}$. If you divide by 3, divide every term of that side.

Q7–Q9 • Open brackets accurately

7. $4(4q + 2) = 50 \Rightarrow 16q + 8 = 50 \Rightarrow \boxed{q = 21/8}$. Multiply *both* terms by 4.

8. $-2(3 - 4x) = 14 \Rightarrow -6 + 8x = 14 \Rightarrow \boxed{x = 5/2}$. Negative times negative is positive.

9. $3(7y + 4) = 9 + 5y \Rightarrow 21y + 12 = 9 + 5y \Rightarrow 16y = -3 \Rightarrow \boxed{y = -3/16}$. Never divide just one term by 3.

Set 4 • Figure it Out, textbook pages 185–189

Q1–Q3 • Blanks, earnings and hidden dots

Q1: (a) $5a - 8 = 37 \Rightarrow \boxed{a = 9}$. (b) $37 - (33 - b) = 35 \Rightarrow \boxed{b = 31}$. (c) $-3(-11 + c) = 45 \Rightarrow \boxed{c = -4}$.

Q2: Let the number of each note be n: $50n + 100n = 750 \Rightarrow \boxed{n = 5}$. Five Rs. 50 and five Rs. 100 notes.

Q3: The figure has **three blobs** and **four visible blue dots**. Let d be dots under one blob. $3d + 4 = 25 \Rightarrow 3d = 21 \Rightarrow \boxed{d = 7}$.

Q4–Q5 • Reverse the number machines

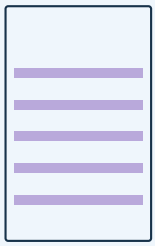
Q4(a): Output is $4(x+3)-5$. For 43, $4(x+3)=48$, $\boxed{x=9}$. For 75, $4(x+3)=80$, $\boxed{x=17}$.
Q4(b): The branches calculate $3x$ and $x+3$; top minus bottom $= 3x-(x+3) = 2x-3$. For 63, $\boxed{x=33}$; for 227, $\boxed{x=115}$.
Q5: Two successive divisions by 3 give output 5: $x/9=5 \Rightarrow \boxed{x=45}$. Two subtractions of 4 give -11: $x-8=-11 \Rightarrow \boxed{x=-3}$.

Q6–Q7 • Taxi fare and a ratio of numbers

Q6: Daily fare Rs. 800 plus Rs. 20 per kilometre. $800+20d=2200 \Rightarrow 20d=1400 \Rightarrow \boxed{70 \text{ km}}$.
Q7: Numbers x and $3x$ total 76. $4x=76 \Rightarrow x=19$. Numbers $\boxed{19 \text{ and } 57}$.

Q8 • Gap between grill rods

Let each empty gap be g cm. The figure shows **two outer borders of 3 cm each**, **five internal bars of 2 cm each**, and **six equal empty gaps**.
$2(3)+5(2)+6g=34 \Rightarrow 6g=18 \Rightarrow \boxed{g=3 \text{ cm}}$.



5 internal rods
6 gaps
Total height 34 cm

Illustration is schematic; the equation uses the dimensions in NCERT's picture.

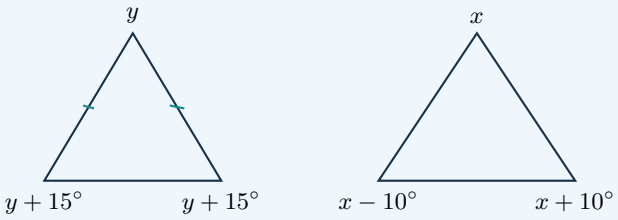
Q9–Q10 • Drinks and linked expressions

Q9: Let a milkshake cost m rupees; juice costs $m-15$. $4(m-15)+7m=600 \Rightarrow 11m=660 \Rightarrow \boxed{m=60}$; juice $= \boxed{45}$ rupees.
Q10: $28p-36=98 \Rightarrow 28p=134$. Then $14p-19=67-19=\boxed{48}$ and $28p-38=134-38=\boxed{96}$.

Q11 • Three solution chains containing mistakes

(a) $6x+9=66 \Rightarrow 6x=57 \Rightarrow \boxed{x=19/2}$. Cannot divide $6x$ by 6 without also dividing the other term.
(b) $14y+24=36 \Rightarrow 7y+12=18 \Rightarrow 7y=6 \Rightarrow \boxed{y=6/7}$. This *is correct*; no error to fix.
(c) $4x-5=9x+8 \Rightarrow -5x=13 \Rightarrow \boxed{x=-13/5}$. Adding 5 to RHS gives 13, not 3.

Q12 • Find angles using 180°



Left (isosceles): $y + 2(y + 15) = 180 \Rightarrow 3y = 150 \Rightarrow y = 50°$. Angles $\boxed{50°, 65°, 65°}$.

Right: $x + (x - 10) + (x + 10) = 180 \Rightarrow 3x = 180 \Rightarrow x = 60°$. Angles $\boxed{60°, 50°, 70°}$.

Q13–Q15 • Four equations, a history problem, a giraffe

Q13: With solution $u = 6$, examples: $u + 1 = 7$, $2u = 12$, $3u - 2 = 16$, $u/2 = 3$.

Q14: First person a, second $2a$, third $6a$, fourth $24a$. Sum $33a = 132 \Rightarrow \boxed{a = 4}$. Amounts: $4, 8, 24, 96$.

Q15: Giraffe height $h = h/2 + 2.5 \Rightarrow h/2 = 2.5 \Rightarrow \boxed{h = 5 \text{ m}}$.

Q16 • Two different stick patterns

Pattern A: a row of squares plus a triangular point. At step n, squares $= n$, sticks $= 3n + 3$ (first step 6 sticks; add 3 per extra square).

(a) At step 11: $\boxed{11 \text{ squares}}$. (b) Sticks: $3(11) + 3 = \boxed{36}$. (c) $3n + 3 = 85 \Rightarrow n = 82/3$: **not possible**. (d) $3n + 3 = 150 \Rightarrow \boxed{n = 49}$.

Pattern B: a stepped zigzag of pairs of squares. At step 1 there are 4 squares and 13 sticks; each new step adds 2 squares and 6 sticks. Hence squares $= 2n + 2$, sticks $= 6n + 7$.

(a) At step 11: $\boxed{24 \text{ squares}}$. (b) Sticks $= 6(11) + 7 = \boxed{73}$. (c) $6n + 7 = 85 \Rightarrow \boxed{n = 13}$. (d) $6n + 7 = 150 \Rightarrow n = 143/6$: **not possible**.

Counting reminder: each new pair in the lower figure shares an existing edge; do not count it twice.

Q17–Q18 • One number and eight equations

Q17: $a + 36 = 10a \Rightarrow \boxed{a = 4}$.

Part	Equation and important step	Solution
(a)	$5(r + 2) = 10 \Rightarrow r + 2 = 2$	$\boxed{r = 0}$
(b)	$-3(u + 2) = 2(u - 1) \Rightarrow -5u = 4$	$\boxed{u = -4/5}$
(c)	$2(7 - 2n) = -6 \Rightarrow 14 - 4n = -6$	$\boxed{n = 5}$
(d)	$2(x - 4) = -16 \Rightarrow x - 4 = -8$	$\boxed{x = -4}$
(e)	$6(x - 1) = 2(x - 1) - 4 \Rightarrow 4(x - 1) = -4$	$\boxed{x = 0}$
(f)	$3 - 7s = 7 - 3s \Rightarrow -4s = 4$	$\boxed{s = -1}$
(g)	$2x + 1 = 9 - 2x \Rightarrow 4x = 8$	$\boxed{x = 2}$
(h)	$10 - 5x = x + 2 \Rightarrow 6x = 8$	$\boxed{x = 4/3}$

Q19 • Follow the equation-maze arrows

Follow the arrow labelled with the solution of the current equation (see textbook p. 189). A valid route is:

$$\boxed{\text{START: } x = 4} \longrightarrow \boxed{x = 3} \longrightarrow \boxed{x = 21} \longrightarrow \boxed{x = 1}$$
$$\longrightarrow \boxed{x = 2} \longrightarrow \boxed{x = 8} \longrightarrow \boxed{x = 13}$$
$$\longrightarrow \boxed{k = 4} \longrightarrow \boxed{x = 4} \longrightarrow \boxed{\text{END}}$$

Check examples: $8x = 20 + 3x \Rightarrow x = 4$; $2x - 9 = -3 \Rightarrow x = 3$; $-2x = -42 \Rightarrow x = 21$; $15 = 19 - 4x \Rightarrow x = 1$. Continue choosing a connected arrow that displays the value just solved; the maze is about both *equations and direction*.

Q20 • Children and donkeys

Let d be donkeys, c children. Heads: $c + d = 28$. Feet: $2c + 4d = 80$. Subtract twice the first equation from the second: $2d = 24 \Rightarrow \boxed{d = 12}$; then $\boxed{c = 16}$. Check $12 \times 4 + 16 \times 2 = 80$.

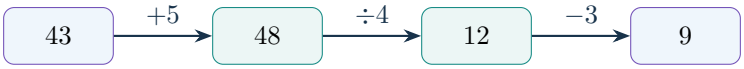
Part B • Understand why equations work

1. A balance remains balanced if both sides change equally.



Remove 3 from both sides, then divide by 2: $x = 5$.

2. Undo operations in reverse order. To solve $4(x + 3) - 5 = 43$, add 5, divide by 4, then subtract 3. The order matters.



Exam strategy: five steps

(1) Define the unknown in words; (2) form one accurate equation; (3) use the same operation on both sides; (4) solve; (5) substitute back and include units. A negative solution is not automatically wrong, but negative *length or number of objects* may be impossible in context.

Part C • Memory tricks and common mistakes

- **Moving across = is shorthand:** "move +7 to the other side" really means subtract 7 *from both sides*.
- **Distribute to every term:** $-2(3 - 4x) = -6 + 8x$, not $-6 - 8x$.
- **Check for special equations:** $5s = 3s$ has solution zero; $x+4 = x+5$ has no solution; $2(x+1) = 2x+2$ holds for every x.
- **Never divide by zero.** From $0x = 0$ we cannot conclude $x = 0$; all values work.
- **Picture puzzles demand careful counting.** Identify shared edges, visible dots, bars and gaps before writing your equation.
- **Angle sum in a triangle:** 180°; equal sides face equal angles.